

Sheet - 8 Functions - I

1.Square root of a number

Given a number A. Return square root of the number if it is perfect square otherwise return -1.

Note: A number is a perfect square if its square root is an integer.

2.Area of Square

You are given a positive integer A denoting the side of a square. You have to calculate the area of the square.

Area of a square having side S is given by $(S * S)$.

3.Area of Circle

You are given a positive integer A denoting the radius of a circle. You have to calculate the area of the circle.

4.Power function

You are given two integers A and B. You have to find the value of A^B .

5.Cube It!

You are given an integer A. You have to find the value of cube of A i.e, A^3 .

6.Volume Of Sphere

You are given a positive integer A denoting the radius of a sphere. You have to calculate the volume of the sphere.

Volume of a sphere having radius R is given by $(4 * \pi * R^3) / 3$.

7.Area Of Ellipse

Given the lengths of semi-major axis A and semi-minor axis B of an ellipse, calculate the Area of the Ellipse.

Area of ellipse having semi-major axis length a and semi-minor axis length b is given by $\pi * a * b$.

8.Sum the Array

Write a program to print sum of elements of the input array A of size N.

9.Square of all numbers

WAP to print square of all numbers from x to y

10.Adding Two numbers

WAP to adding two numbers by using functions.

11. Calculator

WAP to make a calculator by using functions.

12. Pythagoras Theorem

WAP to implement the Pythagoras Theorem in Python using functions.